

Almost everywhere divergence of polynomial interpolation with sublinear excess degree

Abstract

Let X_n be an arbitrary set of n distinct points in $[-1, 1]$, and let r_n be nonnegative integers with $r_n = o(n)$. We prove that there is a real continuous function f such that every sequence of polynomials p_n , of degree at most $n + r_n$, satisfying $p_n = f$ on X_n , has $\limsup_{n \rightarrow \infty} |p_n(x)| = \infty$ for almost every $x \in [-1, 1]$. The exceptional null set may depend on the sequence (p_n) . This strengthens the divergence assertion in Erdős Problem 1152. The proof combines localized functions in Bernstein spaces with uniform asymptotics for weighted Christoffel–Darboux kernels. A finite interpolation construction and the Baire category theorem give a single function valid for every admissible interpolation sequence.

1 Introduction

Write Π_d for the space of real polynomials of degree at most d . All measures of subsets of $[-1, 1]$ below are Lebesgue measures. The uniform norm on this interval is denoted by $\|\cdot\|_\infty$.

Theorem 1. *For each $n \geq 1$, let $X_n \subset [-1, 1]$ consist of n distinct points. Suppose that $r_n \geq 0$ are integers and $r_n/n \rightarrow 0$. There exists $f \in C([-1, 1]; \mathbb{R})$ such that every sequence satisfying*

$$p_n \in \Pi_{n+r_n}, \quad p_n(x) = f(x) \quad (x \in X_n)$$

also satisfies

$$\limsup_{n \rightarrow \infty} |p_n(x)| = \infty \quad \text{for almost every } x \in [-1, 1]. \quad (1.1)$$

This theorem strengthens the divergence assertion in Erdős Problem 1152 [1], which asks whether the interpolants must fail to converge to f almost everywhere. The exceptional null set in (1.1) may depend on the sequence (p_n) . The same conclusion holds for polynomials with complex coefficients, by applying the real statement to their real parts.

The two analytic ingredients are the construction in Olevskii and Ulanovskii [3, Lemma 4.1] and the uniform kernel asymptotics of Kriecherbauer, Schubert, Schöler, and Venker [2, Theorem 1.3 and Theorem 1.8 (bulk universality)]. We first obtain interpolation data that force large values on a fixed fraction of a local interval, uniformly over all admissible polynomial corrections. We then combine finitely many such data sets to rule out common sets of bounded values. The final step is a category argument in $C([-1, 1]; \mathbb{R})$.

2 Localized Bernstein functions and polynomial corrections

For $\sigma > 0$, let B_σ denote the space of entire functions of exponential type at most σ that are bounded on $\mathbb{R}$. Norms of functions in B_σ are taken on $\mathbb{R}$.

Lemma 2. *There are absolute constants $c, C > 0$ with the following property. For every sufficiently large integer ρ , put*

$$R_0 = \rho - \sqrt{\rho}, \quad \gamma = R_0^{-1/9}.$$

For every finite set $\Gamma \subset [-\rho, \rho]$ with $\#\Gamma \leq 2\rho + 1$, there is an entire function g , real on $\mathbb{R}$, such that

$$g \in B_{\pi-4\gamma}, \quad \max_{\Gamma} |g| \leq 1, \quad \|g\|_{\infty} \leq C \log \rho, \quad (2.1)$$

$$g(u_0) \geq c \log \rho \quad \text{for some } |u_0| \leq R_0, \quad (2.2)$$

$$|g(u)| \leq C(|u| - R_0)^{-3} \quad (|u| \geq \rho). \quad (2.3)$$

These functions have uniformly bounded $L^1(\mathbb{R})$ norms and uniformly vanishing integral tails when ρ is fixed. There is also an interval, of length bounded below by an absolute positive constant, contained in $(-\rho, \rho) \setminus \Gamma$, on which $g \geq c' \log \rho$, with an absolute constant $c' > 0$.

Proof. Use the construction of [3, Lemma 4.1], replacing its initial bandwidth $\pi - 4R_0^{-1/9}$ by $\pi - 8R_0^{-1/9}$. Put $w = \pi - 8\gamma$ and

$$\Lambda = \Gamma + 2R_0\mathbb{Z}.$$

Since Γ is finite, Λ is uniformly discrete. Being periodic, its lower and upper uniform densities coincide; write their common value as D . Then

$$D \leq \frac{2\rho + 1}{2R_0} \leq 1 + CR_0^{-1/2}.$$

Write $K_s(\Lambda, B_w)$ for the sampling constant. If Λ is a sampling set for B_w , then [3, Theorem C(i)] gives $D > w/\pi$. Hence $D\Lambda$ has density one and [3, Theorem 1(i)] gives

$$K_s(\Lambda, B_w) = K_s(D\Lambda, B_{w/D}) \geq C \log \frac{\pi}{\pi - w/D} \geq c \log R_0,$$

since, for large ρ ,

$$0 < \pi - \frac{w}{D} = \frac{\pi D - w}{D} \leq C(R_0^{-1/2} + \gamma) \leq C\gamma.$$

If Λ is not a sampling set, its sampling constant is infinite, so the same lower bound is immediate. Thus one can choose $g_1 \in B_w$, bounded by one on Λ , whose norm is a fixed constant times $\log R_0$. Choose a point at which its absolute value is at least half this norm. A translation by a multiple of $2R_0$ places this point at $u_0 \in [-R_0, R_0]$, without changing the bound on Λ . Multiplying by a constant of modulus one and taking the real part gives a function real on $\mathbb{R}$ and positive at u_0 .

Multiply this function by

$$\left(\frac{\sin(\gamma(u - u_0))}{\gamma(u - u_0)} \right)^4.$$

The resulting function has bandwidth at most $\pi - 4\gamma$ and retains the node bound and the value at u_0 . For $|u| \geq \rho$, the bound

$$C \log R_0 \gamma^{-4}(|u| - R_0)^{-4} \leq C(|u| - R_0)^{-3}$$

follows from $\log R_0 R_0^{4/9}/\sqrt{\rho} = O(1)$. This proves (2.1)–(2.3) and the assertions about integrability. Finally, Bernstein's inequality gives $\|g'\|_{\infty} \leq \pi C \log \rho$. Thus the peak contains an interval of fixed positive length on which $g \geq c' \log \rho > 1$. This interval contains no point of Γ and lies inside the original box. $\square$

For a finite set $Y \subset \mathbb{R}$, define

$$P_Y(x) = \prod_{y \in Y} (x - y).$$

Lemma 3. *Let $J_1, \dots, J_K$ be disjoint intervals, in increasing order, none of which meets Y . Suppose that a real polynomial b satisfies*

$$|b| > H, \quad \text{sgn}(b/P_Y)|_{J_j} = (-1)^j. \quad (2.4)$$

If $p = b + P_Y q$ with $\deg q \leq d$, then at least $(K - d - 1)/2$ of these intervals satisfy $|p| > H$ at every point.

Proof. Call J_j a low interval if it contains a point x_j with $|p(x_j)| \leq H$. At such a point the sign of $q(x_j)$ is opposite to that of $b(x_j)/P_Y(x_j)$. If there are B low intervals, the resulting signs alternate at least $2B - K - 1$ times. Indeed, deleting $K - B$ terms from a fully alternating sequence removes at most $2(K - B)$ sign changes. Consequently $2B - K - 1 \leq d$, so

$$B \leq \frac{K + d + 1}{2}. \quad (2.5)$$

The asserted bound follows. $\square$

3 An explicit external field and its polynomial kernel

For a finite positive measure μ , use the logarithmic-potential convention

$$V_\mu(x) = \int \log |x - y| d\mu(y).$$

Define

$$Q(u) = \frac{1}{2}((1 + u) \log(1 + u) + (1 - u) \log(1 - u)), \quad |u| < 1. \quad (3.1)$$

Then $Q(0) = 0$, $Q'(u) = \text{arctanh } u$, and $Q''(u) = (1 - u^2)^{-1}$.

Lemma 4. *For $0 < t < 1$, set $a_t = \sqrt{1 - (1 - t)^2}$. The equilibrium measure of mass t in the external field Q on $[-1, 1]$ has density*

$$\rho_t(u) = \frac{1}{\pi} \arccos \frac{1 - t}{\sqrt{1 - u^2}} \mathbf{1}_{\{|u| < a_t\}}. \quad (3.2)$$

Thus $Q - V_{\rho_t}$ is constant on $[-a_t, a_t]$ and is no smaller elsewhere in $[-1, 1]$.

Proof. Differentiation with respect to t gives

$$\partial_t \rho_t(u) = \frac{\mathbf{1}_{\{|u| < a_t\}}}{\pi \sqrt{a_t^2 - u^2}}. \quad (3.3)$$

This is the arcsine probability density on $[-a_t, a_t]$, so ρ_t has total mass t . Its Cauchy transform can be computed by integrating the transforms of these arcsine measures. For $0 < x < a_t$, its real boundary value is

$$\int_0^{1 - \sqrt{1 - x^2}} \frac{ds}{\sqrt{x^2 - a_s^2}} = \text{arctanh } x = Q'(x). \quad (3.4)$$

One may justify the calculation first in the upper half-plane by Fubini's theorem and then take boundary values. For $a_t < x < 1$, the upper limit in (3.4) is t , and the integral is strictly smaller than $Q'(x)$. Symmetry gives the corresponding statements on the negative half-axis. These identities prove the equilibrium equalities and inequalities. $\square$

Let $0 < t < 1$, and choose $a_t < b < 1$. Let $s \rightarrow \infty$, $m = \lfloor ts \rfloor$, and let Q_s be real on $[-b, b]$ and analytic in a fixed complex neighborhood of that interval. Assume that Q_s is sufficiently close to Q there in the uniform norm. The weight e^{-2sQ_s} , with m orthogonal polynomials, corresponds to the external field $2sQ_s/m$ in [2]. The limiting field $2Q/t$ satisfies hypothesis (GA)₂ of that paper: it is analytic and strictly convex, and its equilibrium support is contained in the interior of $[-b, b]$. Theorem 1.3(b) of [2] therefore gives estimates uniform in the indicated neighborhood of Q .

Let $p_0, \dots, p_{m-1}$ be orthonormal for $e^{-2sQ_s(x)} dx$ on $[-b, b]$, and set

$$K_s(x, z) = e^{-sQ_s(x) - sQ_s(z)} \sum_{j=0}^{m-1} p_j(x)p_j(z), \quad c_s = \frac{s}{2}. \quad (3.5)$$

Put $A = [-1/4, 1/4]$. For $y \in A$ and bounded real u, v , [2, Theorem 1.8 (bulk universality)] gives

$$\frac{K_s(y + u/c_s, y + v/c_s)}{c_s} = \frac{\sin(\pi\beta_{s,y}(u - v))}{\pi(u - v)} + o(1). \quad (3.6)$$

Here $\beta_{s,y}$ is twice the density at y of the equilibrium measure of mass m/s for Q_s , and at $u = v$ the quotient on the right is interpreted by continuity. To match the normalization in [2, Theorem 1.8 (bulk universality)], set

$$V_s = \frac{2s}{m}Q_s,$$

so that $e^{-mV_s} = e^{-2sQ_s}$. If ψ_s is the equilibrium probability density for V_s , then $\rho_s = (m/s)\psi_s$ is the equilibrium density of mass m/s for Q_s , and hence

$$m\psi_s(y) = s\rho_s(y) = c_s\beta_{s,y}.$$

Substituting the local variables $\beta_{s,y}u$ and $\beta_{s,y}v$ in [2, Theorem 1.8 (bulk universality)] gives (3.6). The error is uniform for $y \in A$, bounded u, v , and the indicated analytic neighborhood of Q , by part (b) of that theorem.

We also need estimates away from the diagonal. Choose fixed intervals $J_0 \Subset J_1$ in the common interior of the equilibrium supports, with $A \Subset J_0$. The difference-quotient formula in [2, Theorem 1.3] gives

$$\frac{|K_s(x, z)|}{c_s} \leq \frac{C}{1 + s|x - z|} \quad (x, z \in J_1). \quad (3.7)$$

Near the diagonal, this follows from the diagonal estimate and Cauchy-Schwarz. For z within $O(s^{-1})$ of A , the Airy formulas in the same theorem give

$$\frac{|K_s(x, z)|}{c_s} \leq Cs^{-5/6} \quad (x \in [-b, b] \setminus J_0). \quad (3.8)$$

Indeed, the kernel factors at z are bounded, those near a support endpoint are at most $O(s^{1/6})$, and x and z are separated by a fixed positive distance. The constants in (3.7) and (3.8) are uniform for the external fields under consideration.

4 Local amplification with the external-node weight

Consider a sequence of finite node sets $Y \subset [-1, 1]$, with $N = \#Y$, and fix

$$I = (x_0 - h, x_0 + h) \Subset (-1, 1), \quad s = \#(Y \cap I).$$

In the coordinate $u = (x - x_0)/h$, define

$$W(u) = \prod_{y \in Y \setminus I} (x_0 + hu - y), \quad Q_s(u) = -\frac{1}{s} \log \left| \frac{W(u)}{W(0)} \right|. \quad (4.1)$$

The function Q_s is convex on $(-1, 1)$.

Proposition 5. *Fix $H > 1$, and choose the parameters of Sections 2 and 3 as below. Suppose that $s \rightarrow \infty$, that Q_s lies in the specified analytic neighborhood of Q , and that*

$$\#\{y \in Y : (y - x_0)/h \in A\} \leq \frac{s}{4} + \epsilon_\rho s, \quad \epsilon_\rho = \frac{1}{16\rho}. \quad (4.2)$$

If $d = o(s)$, then, for all sufficiently large s , there are data $v_y \in [-1, 1]$, $y \in Y$, such that every $p \in \Pi_{N+d}$ interpolating these data satisfies

$$|\{x \in I : |p(x)| > H\}| \geq c_H |I|. \quad (4.3)$$

The constant $c_H > 0$ depends only on H .

Proof. We fix the auxiliary parameters in the following order, all before letting $s \rightarrow \infty$. First choose ρ so large that

$$c' \log \rho > 10H,$$

where c' is the constant in Lemma 2, and set $R_0 = \rho - \sqrt{\rho}$ and $\gamma = R_0^{-1/9}$. Choose $t < 1$ so close to one that

$$\inf_{y \in A} 2\pi\rho_t(y) > \pi - 2\gamma.$$

Fix $K = 3$, and then choose $\kappa > 0$ so small that $t + \kappa < 1$ and the Jackson factor (4.8) below is larger than $9/10$ whenever $y \in A$ and $|c_s(x - y)| \leq \rho + 1$, for all sufficiently large s . Choose b with $a_{t+\kappa} < b < 1$, choose $A \Subset J_0 \Subset J_1 \Subset (-b, b)$ inside the common bulk region, and shrink the analytic neighborhood of Q so that the kernel estimates of Section 3 are uniform there, the supports of the equilibrium measures of masses between t and $t + \kappa$ stay a fixed positive distance from $\pm b$, and

$$\inf_{y \in A} \pi\beta_{s,y} > \pi - 3\gamma. \quad (4.4)$$

Set $\eta = 1/10$. By the uniform integral-tail bound in Lemma 2, choose T so that the truncation error in (4.6) is less than $\eta/2$. Next choose $R > \max\{\rho + 1, 2T\}$. Finally choose $D > 2R$ so that

$$2C_\rho \sum_{j=0}^{\infty} \frac{1}{1 + R + jD} \frac{1}{(1 + \kappa(R + jD))^{2K}} < \frac{1}{10}. \quad (4.5)$$

Here C_ρ is the constant in (4.10); first increasing R and then D makes (4.5) hold. All these parameters depend only on H . In this proof, the arguments of K_s , $G_{s,y}$, and $F_{s,y}$ are in the scaled coordinate u .

Partition the interval $c_s A$ into boxes of length 2ρ , discarding incomplete boxes at the ends. There are $s/(8\rho) - O(1)$ boxes. Call a box admissible if it contains at most $2\rho + 1$ scaled nodes. If B is the number of admissible boxes, then

$$\left(\frac{s}{8\rho} - O(1) - B \right) (2\rho + 2) \leq \frac{s}{4} + \epsilon_\rho s.$$

The choice of ϵ_ρ implies $B \geq c_\rho s$.

For an admissible box centered at $c_s y$, apply Lemma 2 to its translated nodes, obtaining g . For a fixed truncation parameter T , define

$$G_{s,y}(x) = \int_{-T}^T g(v) \frac{K_s(x, y + v/c_s)}{c_s} dv. \quad (4.6)$$

This is $e^{-sQ_s(x)}$ times a polynomial of degree at most $m - 1$. By (4.4) and the Paley–Wiener theorem,

$$\int_{\mathbb{R}} g(v) \frac{\sin(\pi \beta_{s,y}(u - v))}{\pi(u - v)} dv = g(u).$$

With η, T, R fixed above, Equation (3.6) implies, for all sufficiently large s ,

$$\sup_{|u| \leq R} |G_{s,y}(y + u/c_s) - g(u)| < \eta. \quad (4.7)$$

The uniform L^1 bound makes this assertion uniform even when g varies with the node configuration.

To obtain summable tails, multiply by an algebraic Jackson factor. Write $x = \cos \theta$, $y = \cos \alpha$, and, with $K = 3$ and κ fixed above, set $k = \lfloor \kappa s / (2K) \rfloor$. Define

$$\begin{aligned} D_k(v) &= \frac{\sin((k + 1/2)v)}{(2k + 1) \sin(v/2)}, \\ L_{s,y}(x) &= D_k(\theta - \alpha)^{2K} + D_k(\theta + \alpha)^{2K}. \end{aligned} \quad (4.8)$$

This expression is even in θ , so it is an algebraic polynomial in x of degree at most $2Kk$. Since $y \in A$, it is bounded on $[-1, 1]$ by $1 + o(1)$. By the choice of κ , it is greater than $9/10$ throughout the scaled neighborhood containing the peak interval of g . Moreover,

$$|L_{s,y}(x)| \leq C(1 + \kappa s|x - y|)^{-2K} + Cs^{-2K}. \quad (4.9)$$

Put $F_{s,y} = G_{s,y}L_{s,y}$. For $x \in J_1$ with $|c_s(x - y)| > 2T$, (3.7) gives

$$\begin{aligned} |F_{s,y}(x)| &\leq C_\rho(1 + |c_s(x - y)|)^{-1}(1 + \kappa|c_s(x - y)|)^{-2K} \\ &\quad + O_\rho(s^{-2K}). \end{aligned} \quad (4.10)$$

For $x \in [-b, b] \setminus J_0$, (3.8) and (4.9) give

$$|F_{s,y}(x)| = O_\rho(s^{-2K-5/6}). \quad (4.11)$$

To treat $b < |x| < 1$, write

$$F_{s,y}(x) = e^{-sQ_s(x)} H_{s,y}(x), \quad \deg H_{s,y} \leq D_s := m - 1 + 2Kk < s.$$

Let σ be the equilibrium measure of mass D_s/s for the field Q_s on $[-b, b]$. Its support remains in a fixed compact subinterval of $(-b, b)$. This follows from the continuity of the support endpoints in [2, Lemma 2.4], after shrinking the external-field neighborhood to cover masses near both t and $t + \kappa$. Let F be the constant value of $Q_s - V_\sigma$ on $\text{supp } \sigma$. Since $\deg H_{s,y} \leq D_s = s\sigma(\mathbb{R})$, the maximum principle applied to $\log |H_{s,y}| - sV_\sigma$ on the complement of the support, including infinity, yields

$$|e^{-sQ_s(x)} H_{s,y}(x)| \leq \left\| e^{-sQ_s} H_{s,y} \right\|_{\text{supp } \sigma} e^{-s(Q_s(x) - V_\sigma(x) - F)}. \quad (4.12)$$

For $x \in \text{supp } \sigma \cap J_0$, the points $y + v/c_s$, $|v| \leq T$, lie in J_1 for all large s . Hence (3.7), the uniform L^1 -bound on g , and $\|L_{s,y}\|_\infty \leq 1 + o(1)$ give

$$|F_{s,y}(x)| \leq C_\rho.$$

For $x \in \text{supp } \sigma \setminus J_0$, the same conclusion follows from (3.8), since $y + v/c_s$ stays within $O(s^{-1})$ of A . Consequently

$$\left\| e^{-sQ_s} H_{s,y} \right\|_{\text{supp } \sigma} = \|F_{s,y}\|_{\text{supp } \sigma} \leq C_\rho \quad (4.13)$$

uniformly in the admissible box and in the allowed external field.

Put $D_s^*(x) = Q_s(x) - V_\sigma(x) - F$. The function Q_s is convex on $(-1, 1)$, while $V_\sigma'' < 0$ off $\text{supp } \sigma$. After shrinking the analytic neighborhood of Q , there is $c_0 > 0$ such that $Q_s'' \geq c_0$ on $[-b, b]$, and hence

$$(D_s^*)'' = Q_s'' - V_\sigma'' \geq c_0 \quad \text{off } \text{supp } \sigma \text{ in } [-b, b].$$

Let α_s and β_s be the leftmost and rightmost points of $\text{supp } \sigma$. The support remains a fixed positive distance from $\pm b$. Since $D_s^* \geq 0$ on $[-b, b]$ and $D_s^* = 0$ on the support, its outward one-sided derivative at α_s and β_s is nonnegative. Integrating the preceding inequality from the support to $\pm b$ gives, for some $c > 0$ independent of s ,

$$D_s^*(-b) \geq c, \quad D_s^*(b) \geq c.$$

Outside $[-b, b]$, strict convexity on the two complementary intervals shows that D_s^* continues to increase toward the endpoints. Therefore (4.12) and (4.13) give

$$|F_{s,y}(x)| = O_\rho(e^{-cs}) \quad (b < |x| < 1). \quad (4.14)$$

Retain admissible boxes whose centers in the $c_s u$ coordinate are separated by at least the fixed distance $D > 2R$. By (4.5), the sum of the tails in (4.10) from all centers outside the R -window is less than $1/10$. One residue class of the box indices retains at least $c_{\rho,D}s$ admissible boxes. At any point there is at most one center within scaled distance R . In this window use (4.7). At nodes of the same box use $|g| \leq 1$; outside the box use (2.3). The remaining terms are controlled by the absolutely summable tails (4.10)–(4.14). The $O_\rho(s^{-2K})$ remainders remain $o(1)$ after summing over all retained boxes. Thus, for every choice $\varepsilon_j \in \{-1, 1\}$,

$$\max_{z \in Y \cap I} \left| \sum_j \varepsilon_j F_{s,y_j} \left(\frac{z - x_0}{h} \right) \right| \leq 2. \quad (4.15)$$

On $(-1, 1)$, the ratio $W(u)/W(0)$ is positive, and hence $e^{-sQ_s(u)} = W(u)/W(0)$. The sum in (4.15) therefore extends to a polynomial in the original variable, vanishes at $Y \setminus I$, and has degree at most

$$N - s + m - 1 + 2Kk \leq N - 1. \quad (4.16)$$

Divide it by two. Its values at all nodes now belong to $[-1, 1]$. On the peak interval belonging to the j -th center, (4.7), the choice of κ , and $c' \log \rho > 10H$ give

$$|F_{s,y_j}| \geq \frac{9}{10}(10H - \eta).$$

The other retained centers contribute at most $1/10 + o(1)$ there by (4.5) and (4.10)–(4.14). Hence, for all sufficiently large s , each retained peak contains a node-free interval of length c/s in the

u -coordinate on which the polynomial after division by two has absolute value greater than $2H$, with sign determined by ε_j . Choose these signs so that the resulting polynomial, denoted by b , satisfies (2.4). Set $v_y = b(y)$. Every interpolant $p \in \Pi_{N+d}$ has the form $p = b + P_Y q$, with $\deg q \leq d$. There are at least $c_{\rho,D}s$ peak intervals, whereas $d = o(s)$. Lemma 3 preserves a fixed fraction of them as intervals on which $|p| > H$ everywhere. Returning to the original variable proves (4.3). $\square$

5 Intervals with small node density

A second local construction requires no external field. Let $I \Subset (-1, 1)$, let Y contain N nodes, and write

$$x = \cos \theta, \quad v = \frac{N\theta}{\pi}.$$

Let $I^\theta \subset (0, \pi)$ be the angular interval corresponding to I . Suppose that

$$\#(Y \cap I) \leq \frac{N|I^\theta|}{2\pi}. \quad (5.1)$$

Choose ρ so that $c' \log \rho > 10H$, and then choose $D > 4\rho$ so that

$$2C \sum_{j=1}^{\infty} (jD - 2\rho)^{-3} < \frac{1}{10}. \quad (5.2)$$

Partition the corresponding v -interval into boxes of length 2ρ , discarding the incomplete boxes at the two ends. Let M be the number of complete boxes and B_{bad} the number containing at least $2\rho + 2$ nodes. The v -interval has length $L = N|I^\theta|/\pi$, so (5.1) gives

$$B_{\text{bad}}(2\rho + 2) \leq \frac{L}{2}, \quad L < 2\rho(M + 2).$$

Thus

$$B_{\text{bad}} < \frac{\rho(M + 2)}{2\rho + 2} \leq \frac{M}{2} + 1.$$

Hence at least $M/2 - 1$ complete boxes contain at most $2\rho + 1$ nodes; in particular, this is at least $M/3$ once $M \geq 6$. Retain such boxes with a fixed separation D , and apply Lemma 2 at their centers a_j . For the resulting functions g_j , define

$$\Phi_{N,j}(\theta) = \sum_{\ell \in \mathbb{Z}} \left[g_j \left(\frac{N\theta}{\pi} - a_j + 2N\ell \right) + g_j \left(-\frac{N\theta}{\pi} - a_j + 2N\ell \right) \right]. \quad (5.3)$$

The series converges uniformly by (2.3). Its Fourier coefficients vanish at all frequencies k for which

$$\frac{\pi|k|}{N} > \pi - 4\gamma,$$

by the Paley–Wiener theorem. It is therefore an even real trigonometric polynomial of degree less than N , and hence an algebraic polynomial of degree less than N in $x = \cos \theta$.

Because I^θ stays away from 0 and π , the reflected centers stay a fixed angular distance from $[0, \pi]$, modulo 2π . Their contribution, and that of every nonzero period of the unreflected term, is $O(N^{-3})$ uniformly on $[0, \pi]$, by (2.3). Summing these errors over $O(N)$ retained boxes gives $o(1)$.

By (2.3) and (5.2), the sum of the principal tails from all other retained boxes is less than $1/10$. For each retained box let

$$B_{N,j}(x) = \Phi_{N,j}(\arccos x).$$

If a node $z = \cos \theta$ lies in the j -th retained box, the principal term of $B_{N,j}(z)$ has absolute value at most one by (2.1); if it lies outside that box, the same term is controlled by (2.3). The reflected and nonzero-period terms contribute only $o(1)$ after summation. Therefore, for every choice of signs $\varepsilon_j \in \{-1, 1\}$ and all sufficiently large N ,

$$\max_{z \in Y} \left| \sum_j \varepsilon_j B_{N,j}(z) \right| \leq 2. \quad (5.4)$$

On the other hand, each retained box contains a node-free interval of a fixed positive length in the v -coordinate on which its principal function is at least $c' \log \rho > 10H$. By the choice of D , on this interval its own principal term dominates all the other terms. After dividing the sum in (5.4) by two, each retained box therefore contains a node-free interval on which the resulting polynomial has absolute value greater than $2H$, with sign prescribed by ε_j . Order these peak intervals from left to right and choose the signs so that

$$\operatorname{sgn}(b/P_Y)|_{J_j} = (-1)^j.$$

The polynomial b has degree less than N , and (5.4) gives $|b(y)| \leq 1$ for every $y \in Y$. Setting the interpolation data to $v_y = b(y)$, any interpolant $p \in \Pi_{N+d}$ has the form

$$p = b + P_Y q, \quad \deg q \leq d.$$

There are $c_{\rho,D} N |I^\theta|$ retained peak intervals, whereas $d = o(N)$. Lemma 3 therefore leaves a fixed positive fraction of them on which $|p| > H$ everywhere.

Each peak interval has length bounded below by c/N in the θ -coordinate. For sufficiently small I about any fixed point of $(-1, 1)$, the maximum and minimum of $\sin \theta$ on I^θ have ratio at most two. Hence the total x -length of the surviving intervals is bounded below by a fixed multiple of $|I|$, and we obtain

$$|\{x \in I : |p(x)| > H\}| \geq c_H |I| \quad (5.5)$$

for all interpolants of degree $N + o(N)$. The constant depends only on H , provided (5.1) and this ratio bound hold. This argument does not require the number of nodes in I to tend to infinity.

6 Local intervals for an arbitrary node array

Pass to a subsequence, with indices still denoted by n , such that

$$\nu_n = \frac{1}{n} \sum_{x \in X_n} \delta_x \implies \nu. \quad (6.1)$$

Only this subsequence will be used in the construction. Removing the nodes belonging to any fixed finite set does not alter the weak limit. Put

$$V = V_\nu, \quad m_0 = \inf_{\operatorname{supp} \nu} V.$$

The function V belongs to $L^1[-1, 1]$, and $V \geq m_0$ throughout $[-1, 1]$. To see the latter assertion off the support, note that V is concave on each bounded complementary interval and has the corresponding endpoint limits. On the two exterior intervals it is monotone toward the support.

6.1 Points above the minimum potential

Fix a finite set $S \subset [-1, 1]$ of nodes whose values have already been specified, and put

$$Y_n = X_n \setminus S, \quad P_n = P_{Y_n}.$$

For $z \in \text{supp } \nu$, choose $z_n \in Y_n$ with $z_n \rightarrow z$. Truncation of the logarithmic kernel gives

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log |P'_n(z_n)| \leq V(z), \quad \frac{1}{n} \log |P_n| \rightarrow V \quad \text{in } L^1[-1, 1]. \quad (6.2)$$

For the first assertion, truncate $\log |x - y|$ from below at $-A$. The contribution of the omitted diagonal term is A/n ; one first takes the weak limit and then lets $A \rightarrow \infty$. For the second, use the continuity of

$$y \mapsto (x \mapsto \log |x - y|)$$

as a map from $[-1, 1]$ into $L^1[-1, 1]$.

Suppose first that $V(z) > -\infty$, and let

$$A_{z,\eta} = \{x : V(x) > V(z) + 3\eta\}, \quad \eta > 0.$$

Outside a subset of $A_{z,\eta}$ whose measure tends to zero, (6.2) implies

$$|\ell_n(x)| \geq e^{\eta n}, \quad \ell_n(x) = \frac{P_n(x)}{(x - z_n)P'_n(z_n)}. \quad (6.3)$$

Specify the value one at z_n , and zero at all other nodes of Y_n . Every polynomial of degree at most $n + r_n$ satisfying these data can be written as

$$p = \ell_n T, \quad T(z_n) = 1, \quad \deg T \leq r_n + \#S + 1 = o(n). \quad (6.4)$$

If $|p| \leq H$ on a subset of $A_{z,\eta}$ of fixed positive measure, (6.3) and Remez's inequality yield

$$|T(z_n)| \leq H e^{-\eta n} C^{r_n + \#S + 1} < 1$$

for all sufficiently large n , a contradiction. Here C depends only on the fixed measure lower bound. Thus the measure of the low set in $A_{z,\eta}$ tends to zero uniformly over all the polynomials in (6.4). At every density point of $A_{z,\eta}$, each sufficiently small fixed interval therefore admits data forcing large values on at least half its length.

Choose countably many z for which $V(z)$ decreases to m_0 , and take positive rational η . The sets $A_{z,\eta}$ cover $\{V > m_0\}$ when the chosen source potentials are finite. If some $V(z) = -\infty$, the first assertion of (6.2) instead gives

$$\frac{1}{n} \log |P'_n(z_n)| \rightarrow -\infty.$$

On each set $\{V \geq -A\}$, the same argument gives (6.3) for any fixed positive exponential rate, outside a set of vanishing measure. Letting $A \rightarrow \infty$ covers almost all of $[-1, 1]$. In particular, if $m_0 = -\infty$, this subsection supplies local intervals at almost every point.

6.2 The minimum-potential set

Suppose that $m_0 > -\infty$, and let

$$E = \{x : V(x) = m_0\}.$$

By the differentiation theorem for measures, ν has a finite Lebesgue density at almost every point. At points where this density is zero, all sufficiently small intervals satisfy the strict limiting form of (5.1). Fixing such an interval before taking $n \rightarrow \infty$, and choosing endpoints that are not atoms of ν , allows Section 5 to be applied.

Now let x_0 be a density point of E and a Lebesgue point of the absolutely continuous part of ν , with finite positive density w ; assume also that the singular part has density zero there. These conditions hold at almost every point of E where the density of ν is positive. Put

$$I_h = (x_0 - h, x_0 + h), \quad a_h = \nu(I_h) \sim 2wh,$$

and let μ_h be the image of $\nu|_{I_h}/a_h$ under $x \mapsto (x - x_0)/h$. Then $\mu_h \Rightarrow du/2$ on $[-1, 1]$, so

$$V_{\mu_h} \longrightarrow Q - 1 \quad \text{in } L^1(-1, 1). \quad (6.5)$$

Define the normalized outer potential

$$A_h(u) = -\frac{1}{a_h} \int_{\mathbb{R} \setminus I_h} \log |x_0 + hu - y| d\nu(y).$$

For u corresponding to $E \cap I_h$, the equality $V = m_0$ gives

$$A_h(u) + \frac{m_0}{a_h} - \log h = V_{\mu_h}(u). \quad (6.6)$$

The set of such u occupies a fraction tending to one of $(-1, 1)$. The left side of (6.6) is convex. Consequently (6.5) and (6.6) imply convergence in measure to $Q - 1$, and hence uniform convergence on every compact subinterval. Subtracting the value at zero yields

$$A_h(u) - A_h(0) \longrightarrow Q(u) \quad \text{locally uniformly on } (-1, 1). \quad (6.7)$$

We recall the convexity fact used here. If finite convex functions on an open interval converge in measure to a continuous finite convex function, they converge uniformly on compact subintervals. Indeed, from any subsequence one can extract an almost everywhere convergent subsequence. Convergence points on either side of a given compact interval bound the secant slopes. Convexity then gives equicontinuity on that interval, and the limit is fixed by convergence in measure.

The convergence in (6.7) also holds in a fixed complex neighborhood of each compact subinterval of $(-1, 1)$. Fix $K \Subset (-1, 1)$, and choose a compact interval K_1 whose interior contains $K \cup \{0\}$. Let U be a simply connected complex neighborhood of K_1 whose closure stays a positive distance from $(-\infty, -1] \cup [1, \infty)$. Write $\xi = (y - x_0)/h$; for $y \notin I_h$ one has $|\xi| \geq 1$. Define $\tilde{A}_h$ on U by $\tilde{A}_h(0) = 0$ and

$$\tilde{A}_h'(z) = -\frac{h}{a_h} \int_{\mathbb{R} \setminus I_h} \frac{d\nu(y)}{x_0 + hz - y}.$$

For real $u \in K_1$, integration along the real segment gives $\tilde{A}_h(u) = A_h(u) - A_h(0)$. Moreover,

$$\tilde{A}_h''(z) = \frac{1}{a_h} \int_{\mathbb{R} \setminus I_h} \frac{d\nu(y)}{(z - \xi)^2},$$

so $A_h''(u) = \tilde{A}_h''(u) > 0$ on the real interval.

For sufficiently small fixed $\ell > 0$, the interval $[u - \ell, u + \ell]$ stays in a fixed compact subinterval of $(-1, 1)$ for every $u \in K_1$. Since $|\xi| \geq 1$, the kernels $(u - \xi)^{-2}$ are uniformly comparable on such intervals. Hence

$$A_h''(u) \leq C_\ell \int_{u-\ell}^{u+\ell} A_h''(v) dv = C_\ell (A_h'(u + \ell) - A_h'(u - \ell)).$$

The local uniform convergence in (6.7) and convexity give uniform interior bounds for A_h' , and hence a uniform bound for A_h'' on K_1 . If z lies in a sufficiently thin complex neighborhood of K_1 , then

$$|z - \xi|^{-2} \leq |\operatorname{Re} z - \xi|^{-2},$$

and the same bound holds for $|\tilde{A}_h''(z)|$. Together with $\tilde{A}_h(0) = 0$ and the uniform interior bound on $A_h'(0)$, this gives a normal family on a fixed complex neighborhood of K_1 . Every subsequential limit agrees with Q on the real interval by (6.7), hence equals Q by the identity theorem. Thus $\tilde{A}_h \rightarrow Q$ locally uniformly near K .

Apply this with $K = [-b, b]$. Choose h small enough that $\tilde{A}_h$ belongs to the analytic neighborhood required in Proposition 5 and

$$\mu_h(A) < \frac{1}{4} + \frac{\epsilon_\rho}{2}.$$

This is possible because $\mu_h \Rightarrow du/2$ and $(du/2)(A) = 1/4$. Choose h so that the endpoints of I_h and the preimages of the endpoints of A are not atoms of ν , and keep it fixed. For the fixed finite set S above, put

$$Y_n = X_n \setminus S, \quad s_n = \#(Y_n \cap I_h).$$

Weak convergence then gives

$$\frac{s_n}{n} \rightarrow a_h > 0, \quad \frac{\#\{y \in Y_n : (y - x_0)/h \in A\}}{s_n} \rightarrow \mu_h(A) < \frac{1}{4} + \frac{\epsilon_\rho}{2}.$$

In particular, (4.2) holds for all sufficiently large n . Let $\mathcal{Q}_n$ denote the normalized analytic continuation of the discrete field (4.1), so that $\mathcal{Q}_n(0) = 0$ and

$$\mathcal{Q}_n'(z) = -\frac{h}{s_n} \sum_{y \in Y_n \setminus I_h} \frac{1}{x_0 + hz - y}.$$

On the fixed complex neighborhood used above, the summands are uniformly bounded and equicontinuous as functions of the outer node. Since $n/s_n \rightarrow 1/a_h$, weak convergence of the empirical measures gives $\mathcal{Q}_n' \rightarrow \tilde{A}_h'$ locally uniformly. Integrating from zero yields $\mathcal{Q}_n \rightarrow \tilde{A}_h = A_h - A_h(0)$ locally uniformly in the same neighborhood. If $N_n = \#Y_n$, then the allowed degree $n + r_n$ is $N_n + d_n$, where

$$d_n = r_n + \#(X_n \cap S) = o(s_n).$$

Proposition 5 therefore applies.

Combining the two subsections, and decreasing c_H if necessary, we have proved the following local conclusion. For almost every $x \in (-1, 1)$, there are arbitrarily small intervals I containing x such that, after any fixed finite set of nodes has been removed, every sufficiently late row of the subsequence admits data bounded by one on the remaining nodes and forcing

$$|\{u \in I : |p_n(u)| > H\}| \geq c_H |I|$$

for every interpolant of degree at most $n + r_n$. The constant $c_H > 0$ depends only on H .

7 A finite construction for common low sets

Proposition 6. *Fix $H > 1$, $\delta > 0$, and $N_0 \geq 1$. There are finitely many indices $n_1, \dots, n_k \geq N_0$ and a function $g \in C([-1, 1]; \mathbb{R})$, with $\|g\|_\infty \leq 1$, such that every choice of polynomials*

$$p_j \in \Pi_{n_j+r_{n_j}}, \quad p_j = g \text{ on } X_{n_j},$$

satisfies

$$\left| \bigcap_{j=1}^k \{x : |p_j(x)| \leq H\} \right| < \delta. \quad (7.1)$$

Proof. For any interval $J \subset (-1, 1)$, the intervals furnished by Section 6 form a Vitali cover of almost all of J . Choose finitely many disjoint such intervals whose total length is at least $|J|/2$. For each interval, choose a sufficiently late new row of the fixed subsequence, assigning values only at nodes whose values have not yet been specified. For all choices of the resulting interpolating polynomials, the union of their sets of values exceeding H occupies at least

$$c_*|J|, \quad c_* = \frac{c_H}{2} > 0, \quad (7.2)$$

inside J .

To justify the compatibility of these assignments, let S be the finite set of previously assigned nodes. At a new row n , use only $Y_n = X_n \setminus S$ in the local construction. If b is its polynomial of degree at most $\#Y_n - 1$, every polynomial interpolating all the assigned values on X_n has the form

$$p = b + P_{Y_n}q, \quad \deg q \leq r_n + \#(X_n \cap S) \leq r_n + \#S.$$

Since $\#S$ is fixed before n is chosen, the additional degree is still $o(n)$, so the local conclusion remains applicable.

Assume $0 < \delta \leq 2$, since larger δ is immediate. Suppose that finitely many rows have already been chosen, with degree bounds $d_1, \dots, d_k$. For every admissible choice of their polynomials, the common low set

$$L = \bigcap_{j=1}^k \{x : |p_j(x)| \leq H\}$$

has at most $2 + 2 \sum_j d_j$ boundary points in $[-1, 1]$. A constant polynomial equal to H or $-H$ introduces no additional boundary points. Choose a fixed partition of $[-1, 1]$ into sufficiently short intervals that the cells meeting these boundary points have total length less than $\delta/4$. The same partition works for every admissible choice of the polynomials, since the boundary bound depends only on their degrees.

In each cell, carry out the construction giving (7.2), using new rows and assigning only new node values. If $|L| \geq \delta$, the cells contained entirely in L have total length at least

$$|L| - \delta/4 \geq 3|L|/4.$$

For the new common low set this gives

$$|L_{\text{new}}| \leq (1 - 3c_*/4)|L| \quad \text{whenever } |L| \geq \delta. \quad (7.3)$$

Starting from $L = [-1, 1]$, perform a finite number q of batches, where

$$2(1 - 3c_*/4)^q < \delta.$$

At each batch the new grid is chosen using the degree bounds of the rows already selected. Equation (7.3) shows that the final common low set has measure less than δ , uniformly over all admissible polynomial choices. All rows may be chosen beyond N_0 .

The assigned values define a function on a finite union of nodes, with absolute value at most one. Extend it by linear interpolation between consecutive nodes and by constants to the endpoints of $[-1, 1]$. The resulting continuous function g has $\|g\|_\infty \leq 1$ and satisfies (7.1). $\square$

8 Proof of Theorem 1

Work in the real Banach space $C([-1, 1]; \mathbb{R})$. For positive integers M, N, k , let $\mathcal{F}_{M,N,k}$ consist of the functions f for which there exist polynomials

$$p_n \in \Pi_{n+r_n}, \quad p_n = f \text{ on } X_n,$$

and a measurable set $E \subset [-1, 1]$, with $|E| \geq 1/k$, such that

$$|p_n(x)| \leq M \quad (x \in E, n \geq N).$$

Lemma 7. *Each set $\mathcal{F}_{M,N,k}$ is closed.*

Proof. Let $f_j \in \mathcal{F}_{M,N,k}$ converge uniformly to f . Choose corresponding polynomials $p_{j,n}$ and sets E_j . For each fixed $n \geq N$, Remez's inequality bounds $\|p_{j,n}\|_\infty$ uniformly in j , because $|E_j| \geq 1/k$ and $|p_{j,n}| \leq M$ there. Since the degree is fixed, this also bounds all coefficients. A diagonal subsequence therefore gives polynomials p_n with

$$p_{j,n} \longrightarrow p_n \quad \text{uniformly on } [-1, 1]$$

for every $n \geq N$. The interpolation conditions pass to the limit.

Since $L^1[-1, 1]$ is separable, after a further subsequence,

$$\mathbf{1}_{E_j} \xrightarrow{*} w \quad \text{in } L^\infty[-1, 1], \quad 0 \leq w \leq 1, \quad \int w \geq 1/k.$$

For fixed $n \geq N$ and $\eta > 0$, uniform convergence gives $E_j \cap \{|p_n| \geq M + \eta\} = \emptyset$ for all sufficiently large j . Weak-* convergence therefore gives $w = 0$ almost everywhere on this set. Letting $\eta \downarrow 0$, w is supported, up to a null set, on

$$E = \bigcap_{n \geq N} \{|p_n| \leq M\}.$$

It follows that $|E| \geq \int w \geq 1/k$. For the finitely many indices $n < N$, choose any interpolating polynomials of degree at most $n - 1$. Hence $f \in \mathcal{F}_{M,N,k}$. $\square$

Lemma 8. *Each set $\mathcal{F}_{M,N,k}$ has empty interior.*

Proof. Let an open ball in $C([-1, 1]; \mathbb{R})$ be given. By polynomial density, it contains a polynomial h . Choose $\varepsilon > 0$ such that

$$h + \varepsilon g$$

belongs to the ball whenever $\|g\|_\infty \leq 1$. Apply Proposition 6 with

$$H > \max \left\{ 1, \frac{M + \|h\|_\infty}{\varepsilon} \right\}, \quad \delta = \frac{1}{k},$$

and require all its indices to be at least N and large enough that $n + r_n \geq \deg h$. Let g be the resulting function, and put $f = h + \varepsilon g$.

For every polynomial p_n interpolating f at a selected row,

$$q_n = \frac{p_n - h}{\varepsilon}$$

has degree at most $n + r_n$ and interpolates g on X_n . Moreover, $|p_n| \leq M$ implies $|q_n| < H$. Proposition 6 therefore excludes a common set of measure at least $1/k$ on which all the selected p_n are bounded by M . Thus $f \notin \mathcal{F}_{M,N,k}$, proving that this set contains no open ball. $\square$

Proof of Theorem 1. By Lemmas 7 and 8 and the Baire category theorem, choose

$$f \notin \bigcup_{M,N,k \geq 1} \mathcal{F}_{M,N,k}.$$

Suppose that an admissible interpolation sequence (p_n) satisfied $\limsup_{n \rightarrow \infty} |p_n(x)| < \infty$ for x in a set of positive measure. The identity

$$\{x : \limsup_{n \rightarrow \infty} |p_n(x)| < \infty\} = \bigcup_{M,N \geq 1} \bigcap_{n \geq N} \{x : |p_n(x)| \leq M\}$$

would imply that one of the common low sets on the right has positive measure, and therefore measure at least $1/k$ for some integer k . This would put f in $\mathcal{F}_{M,N,k}$, a contradiction. Equation (1.1) follows. $\square$

References

- [1] T. F. Bloom, *Erdős Problem #1152*, <https://www.erdosproblems.com/1152>. Accessed 4 September 2026.
- [2] T. Kriecherbauer, K. Schubert, K. Schöler, and M. Venker, Global asymptotics for the Christoffel–Darboux kernel of random matrix theory, *Markov Processes and Related Fields* **21** (2015), 639–694. arXiv:1401.6772.
- [3] A. Olevskii and A. Ulanovskii, On irregular sampling and interpolation in Bernstein spaces, *Proceedings of the Steklov Institute of Mathematics* **303** (2018), 178–192.